

Haein Seo

2004.09.18

angela040918@duksung.ac.kr

Tel: 010-8378-1454



EDUCATION

Duksung Women's University, Seoul, Republic of Korea

Mar. 2024 – Expected Feb. 2028

B.S., Department of Data Science, Duksung Women's University (GPA: 3.84/4.5, Major: 3.94/4.5)

Relevant Coursework: Introduction to Machine Learning (A+), Introduction to Deep Learning (A+), Big Data Visualization (A+), Data Processing (A+), Big Data Collection & Information Retrieval (A+), Introduction to Big Data (A+)

RESEARCH INTERESTS

- Computer Vision, 3D Scene Understanding, Segmentation, RGB-D Fusion, Video Understanding
- Multimodal Learning, VLM, VLA, Document Understanding, MLLM, Generative AI
- LLM, RAG, Knowledge Graph, Ontology

EXPERIENCE AND TECHNICAL SKILLS

DMKD Lab, Duksung Women's University

Seoul, Korea

Undergraduate Research Assistant

Jan. 2025 – Present

- Participated in an ETRI-commissioned project developing an AI-based image editing software tool
- Contributed to a National Research Foundation of Korea (NRF)-funded research project
- Attended and presented at academic conferences
- Authored and submitted a journal manuscript
- Conducted research, model implementation, paper reviews, and lab seminars

AWARDS AND HONORS

- Academic Excellence Scholarship
1st in Department (1st Semester 2024 / GPA: 4.25) - Duksung Women's University, 2024 2024
Top 10% (2nd Semester 2024 / GPA: 4.17) - Duksung Women's University, 2024
- 3rd Place, 2025 Samsung AI Challenge: Visually-rich Document Understanding
Samsung, Korea, Oct. 2025 2025
- Minister's Award, Ministry of Science and ICT (Outreach SW Education Volunteer Group)
Korea, Oct. 2025 2025
- Best Paper Award, ACK 2025 (Annual Conference of KIPS)
Korea Information Processing Society, Nov. 2025 2025
- Grand Prize, Department of Data Science Academic Festival
Duksung Women's University, Korea, Nov. 2025 2025
- Gold Prize, SNU FoodTech Commercialization Competition
Seoul National University, Korea, Aug. 2026 2026

Scholarships

- KT Digital Talent Scholarship 2026

INTERPERSONAL ACTIVITIES

- SW Education Mentor, Shinsegae I&C & KOFAC Coding Programs 2024
Shinsegae I&C / Korea Foundation for the Advancement of Science and Creativity
- Cooperation Team, University Career Supporters 2024-2025
Career Development Center, Duksung Women's University, Korea
- Team Lead, Outreach SW Education Volunteer Group 2025
Code Club, Korea
- Administrative Dept., Hanbok Club 2025
Duksung Women's University, Korea
- AI Team, KT Social Problem-Solving Project 2026
KT

PUBLICATIONS AND POSTERS

• Publications

- 1) Haein Seo, Eun-Chong Lee and Jehyeok Rew. "Pedestrian Risk Area Analysis Using Street View-based Object Detection and Urban Spatial Information" *The Transactions of the Korea Information Processing Society* 15, no.8 (2026) : 681-695.
- 2) Haein Seo and Jehyeok Rew. "An Explainable Breast Cancer Diagnosis Framework using SAM-based Lesion Segmentation and VLM-based Semantic Interpretation" *Journal of Platform Technology* 14, no.3 (2026) : 59-70.doi: 10.23023/JPT.2026.14.3.059
- 3) H.-I. Seo, E.-C. Lee, and J. Rew, "Pedestrian safety monitoring method using street view object detection and urban spatial information analysis," in *Proc. Annu. Conf. KIPS*, 2025, pp. 520–521.
- 4) H.-I. Seo, S.-M. Seo, Y.-J. Lee, S.-H. Kim, and J. Moon, "Hybrid clustering-prediction model and XAI analysis for severity stratification in malignant breast cancer patients," in *Proc. ICT Platform Soc. Fall Conf.*, Seongnam, South Korea, 2025, pp. 66–70.
- 5) K. Kim, M. Lee, H. Seo, and J. Rew, "CoDU: Multi-Stage Approach for Complex Document Structure Understanding," in *Proc. Korea Database Conf. (KDBC)*, Gyeongju, Republic of Korea, Nov. 2025, pp. 167–170.
- 6) H. Seo and J. Rew, "PRIME-K: A Multimodal Paper-to-Code Reconstruction Framework for Keystroke Dynamics Research," in *Proc. Korea Computer Congress (KCC)*, Republic of Korea, 2026, pp. 522–524.

CERTIFICATE AND LICENSE

- Google Analytics Certification, Google Sep. 2024